

mRNA *In vitro* Synthesis Solutions

Template Generation • Template Linearization

In vitro Transcription • RNA Analysis



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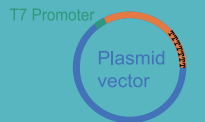
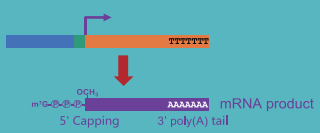
Products manufactured in compliance with GMP regulations can meet diverse customer requirements.

GMP Raw Material Quality Standards

Quality Control	Quality Standard	Detection Method
Appearance	Colorless clear liquid	Visual Check
Protein Purity	≥ 95%	SDS-PAGE
Non-specific Endonuclease Activity	Supercoiled plasmid conformational change < 20%	20 U enzyme was incubated with supercoiled plasmid at 37°C for 4 hours
DNase Activity	No dsDNA degradation observed	20 U enzyme was incubated with dsDNA fragments at 37°C for 16 hours
RNase Activity	RNA degradation ≤ 10%	20 U enzyme was incubated with RNA at 37°C for 1 hour
Host Cell DNA	< 10 copies/20 U	ChP(2025) Volume IV, Detection of Exogenous DNA Residues, Method 3 (General Chapter 3407)
Host Cell Protein	< 50 ppm	ChP(2025) Volume IV, Measurement of E. coli Host Cell Proteins (General Chapter 3412)
Bacterial Endotoxin	< 10 EU/mg	ChP(2025) Volume IV, Bacterial Endotoxins Test, Method 1 (General Chapter 1143)
Mycoplasma	Negative	Mycoplasma test kit
Heavy Metals	< 10 ppm	ChP(2025) Volume IV, Limit Test for Heavy Metals, Method I (General Chapter 0821)
Microbial Limit	The total aerobic microbial count is below 5 cfu/ml, and the total combined yeasts/molds count is below 5 cfu/ml.	ChP(2025) Volume IV, Microbial Limit Tests for Non-Sterile Products: Microbial Enumeration Method (General Chapter 1105)

Note: The above information is given as examples of Bsal, GMP Grade.

mRNA Vaccine Manufacturing Process

Stage	Schematic Diagram
1. Acquisition of antigen genes	
2. Construction and amplification of Recombinant plasmid	
3. Linearization of circular plasmid template	
4. <i>In vitro</i> transcription (co-transcription with cap analogue)	
5. Lipid nanoparticle encapsulation	

BioExacta ENZYMES offer a wide range of GMP and non-GMP grade raw materials, assisting in the research and production of mRNA vaccines.

mRNA Synthesis Workflow Example & Available Products

1	2	3	4	5
Template Generation	Template Linearization	<i>In vitro</i> Transcription	RNA Capping*and Tailing	RNA Analysis
Restriction Endonucleases	Bsal, GMP Grade 	T7 High Yield RNA Synthesis Kit	Cap1 analogue AG †	RNase A
2× S705 HiFi Master Mix	XbaI, GMP Grade 	T7 RNA Polymerase 	Poly(A) RNA Polymerase	RNase T1
T4 DNA Ligase (Fast)	XhoI, GMP Grade 	High T7 RNA Polymerase		Thermostable RNase T1
DNA Assembly Mix Ultra/Mono	NheI, GMP Grade 	RNase Inhibitor, Murine 		RNase H
Golden Gate Assembly Kit (BsmBI)	BspQI, GMP Grade 	DNase I, RNase-free		Thermostable RNase H
Golden Gate Assembly Kit (BpiI)	Esp3I (BsmBI), ADCF †	DNase I-ST, RNase-free 		MazF †
2× Taq PCR Aurora Mix	EcoRI, ADCF †	Pyrophosphatase, Inorganic (Yeast) 		RNase GG †
phi29 II DNA polymerase	Bsal ELISA Kit	Pseudo-UTP		
TeIN Protelomerase	XhoI ELISA Kit	N ¹ -Methyl-Pseudo-UTP		
dNTPs	XbaI ELISA Kit	NTPs		
	BsmBI ELISA Kit			

*Accurately controlling poly(A) tail length using enzymatic methods is challenging, which is why the poly(A) sequence is typically incorporated directly into the plasmid template. † Available on request.

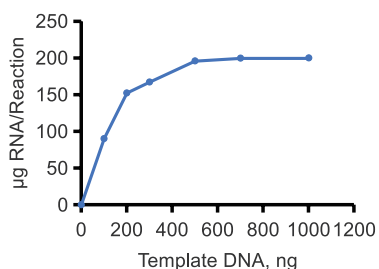
In vitro Transcription

The T7 High Yield RNA Synthesis Kit is capable of synthesizing long or short RNA transcripts in large quantities from DNA templates containing a T7 promoter. The kit's T7 Enzyme Mix is optimized with RNase inhibitor and inorganic pyrophosphatase, ensuring highly efficient and robust RNA synthesis. The synthesized RNA is applicable for a wide range of downstream applications, including *in vitro* translation, RNA structure and function studies, RNase protection assays, probe hybridization, RNA interference, and more.

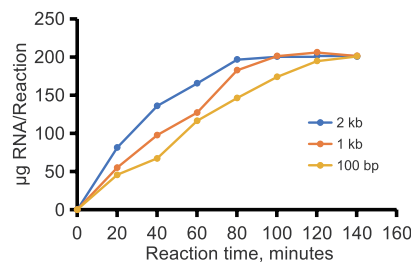
Product Feature

- High yield, 1 µg of linearized double-stranded DNA template can yield at least 150 µg of RNA.
- Fast rate, 80 minutes will be sufficient to obtain the required amount of the product.
- Wide adaptability to template lengths (100 bp to 9 kb), suitable for preparing mRNA, CRISPR-required sgRNA, and other applications.
- Optimized reaction system, compatible with modified NTPs and cap analogs.

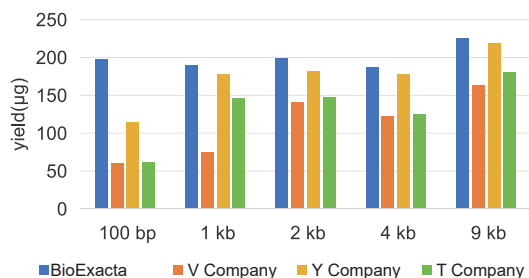
Performance Demonstration



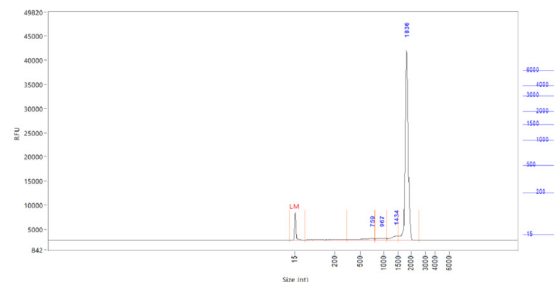
The relationship between amount of template and product.



The relationship between reaction time of different templates length and product output.



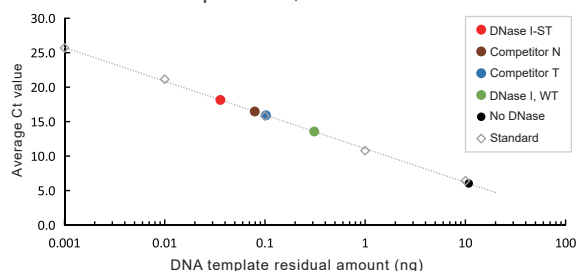
The relationship between different template lengths and the products output.



As tested by customers, when using cap analogues for co-transcription, the integrity of the capped mRNA exceeds 85%.

DNase I-ST, GMP Grade—Through protein design and modification, BioExacta ENZYMES has developed a new salt-tolerant DNase I based on the wild-type DNase I. This product features:

- Significantly enhanced DNA affinity, enabling higher digestion efficiency for low-concentration DNA.
- High salt tolerance, which can efficiently degrade DNA even in solutions containing 300 mM salt.
- Recombinant expression, with no RNase contamination.
- Especially suitable for removing DNA templates after IVT reactions.



	Average Ct value	Residual DNA (ng)	Residual DNA ratio	DNA removal fold
DNase I-ST	18.1	0.036	0.332%	~300
Competitor N	16.5	0.079	0.732%	~130
Competitor T	15.9	0.102	0.948%	~100
DNase I, WT	13.6	0.311	2.875%	~30
No DNase	6.0	10.805	100%	-

As shown in the image, *in vitro* transcription (IVT) reactions using different competitor systems—all with high salt concentrations—were performed using linearized plasmid DNA as template. After transcription, 2 U of DNase I were added. The RNA products were purified and recovered, and residual DNA was quantified by qPCR. The results demonstrate that BioExacta ENZYMES DNase I-ST achieved significantly higher DNA removal efficiency than the wild-type DNase I and competing products.

Template Generation

When amplifying the target fragment, it is essential to select a high-fidelity DNA polymerase to ensure the accuracy of the template. BioExacta ENZYMES has launched the 2×S705 HiFi Master Mix, a high-fidelity PCR premix containing S705 High-Fidelity DNA Polymerase. This polymerase offers 70-fold higher fidelity than Taq polymerase and 5-fold faster amplification than Pfu polymerase, meeting the requirements for accurate template amplification.

BioExacta ENZYMES offers a comprehensive catalog including over a hundred restriction endonucleases, thermostable reverse transcriptase, T4 DNA Ligase (Fast), and Gibson assembly cloning kits, effectively addressing the needs for plasmid template construction.

Template Linearization

When utilizing plasmid DNA as an *in vitro* transcription template, it's crucial to select appropriate restriction endonucleases to ensure complete digestion and linearization of the circular plasmid. BioExacta ENZYMES offers a diverse range of GMP-grade enzymes. We guarantee full traceability throughout our entire production process, including all raw materials and excipients. Our enzymes are completely free of antibiotics and any animal-derived components. Furthermore, we rigorously control process-related impurities such as host proteins, exogenous DNA, non-specific endonucleases, DNase, and RNase, as well as microbial limits and bacterial endotoxins. This stringent quality control meets the demanding raw material and excipient requirements for critical applications like vaccine and drug production and cell therapy.

Meanwhile, BioExacta ENZYMES provides GMP-grade restriction endonucleases customization to meet users' personalized needs for plasmid templates.

To address the issue of restriction endonuclease residue after plasmid template linearization, BioExacta ENZYMES pioneered the industry's first series of detection kits for restriction endonuclease residues. Currently, ELISA kits for individual enzymes including BsaI, XhoI, XbaI, and BsmBI are available, with new products to be gradually launched according to customer demands.

RNA Analysis

When performing *in vitro* transcription with both capping and tailing to generate complete mRNA, it's crucial to analyze the quality of the RNA product. To address this need, BioExacta ENZYMES offers a diverse array of RNase. These enzymes provide versatile options for RNA secondary structure analysis, capping rate evaluation, and comprehensive integrity assessment.

Product Name	Reaction Substrate	Recognition Sequence	Application
RNase A	Primarily single-stranded RNA	C/U	Preparation of plasmids and genomic DNA Remove RNA from recombinant protein preparations RNA In combination with RNase T1 Identify single-base mutations in DNA or RNA
RNase T1/ Thermostable RNase T1	Single-stranded RNA	G	RNA sequencing Analysis of RNA secondary structure Combined with RNase A to remove RNA mRNA quality analysis
RNase H/ Thermostable RNase H	The RNA chain in the RNA-DNA hybrid molecule	Positional preference	Analysis of mRNA cap addition efficiency Remove mRNA before synthesizing the second strand of cDNA Removal of the poly(A) sequence from mRNA after oligo(dT) hybridization

Ordering Information

Classification	REF No.	Name	Specs
Template Generation	BE02002S	M-MLV GIII Reverse Transcriptase	10000 U
	BE01020S/M/L	2× S705 HiFi Master Mix	1 ml/5×1 ml/20×1 ml
	BE05001S	T4 DNA Ligase (Fast)	1000 U
	BE05013S	DNA Assembly Mix Ultra	50 rxns
	BE05025S/M	DNA Assembly Mix Mono	50 rxns/250 rxns
	BE05027S	Golden Gate Assembly Kit (BsmBI)	50 rxns

Ordering Information

Classification	REF No.	Name	Specs
Template Generation	BE05029S	Golden Gate Assembly Kit (Bpil)	50 rxns
	BE01023M/L	2× Taq PCR Aurora Mix	5×1 ml/100×1 ml
	BE01025S/M	phi29 II DNA Polymerase	500 U/2500 U
	BE05024S/M	TelN Protelomerase	250 U/1000 U
	BE01015S	dNTPs (10 mM each)	1 ml
	BE01010S	dATP (100 mM)	1 ml
	BE01011S	dTTP (100 mM)	1 ml
	BE01012S	dCTP (100 mM)	1 ml
	BE01013S	dGTP (100 mM)	1 ml
	BE01014S	dUTP (100 mM)	1 ml
Template Linearization	BE04103SG	Bsal, GMP Grade	20000 U
	BE04106SG	XbaI, GMP Grade	20000 U
	BE04104SG	BspQI, GMP Grade	10000 U
	BE04107SG	XhoI, GMP Grade	20000 U
	BE04105SG	NheI, GMP Grade	20000 U
	BE10002S	Bsal ELISA Kit	96 T
	BE10005S	XhoI ELISA Kit	96 T
	BE10004S	XbaI ELISA Kit	96 T
In vitro Transcription	BE09018S	T7 High Yield RNA Synthesis Kit	50 rxns
	BE09001SG/MG	T7 RNA Polymerase, GMP Grade	20000 U/200000 U
	BE09002S/M	T7 RNA Polymerase	5000 U/25000 U
	BE09003S/M	High T7 RNA Polymerase	5000 U/25000 U
	BE09004S	ATP (100 mM)	1 ml
	BE09005S	CTP (100 mM)	1 ml
	BE09006S	GTP (100 mM)	1 ml
	BE09007S	UTP (100 mM)	1 ml
	BE09008S	NTP Bundle (25 mM each)	1 ml
	BE09009S	N1-Methyl-Pseudo-UTP (100 mM)	100 µl
	BE09010S	Pseudo-UTP (100 mM)	200 µl
	BE09011SG	Murine RNase Inhibitor, GMP Grade	40000 U
	BE09012S	RNase Inhibitor, Murine	10000 U
	BE09013SG	DNase I-ST, GMP Grade	2000 U
	BE09014S	DNase I-ST, RNase-free	1000 U
	BE09015S	DNase I-ST, RNase-free (Lyophilized)	1000 U
	BE09016SG	Pyrophosphatase, Inorganic (Yeast, GMP Grade)	100 U
	BE09017S/M	Pyrophosphatase, Inorganic (Yeast)	10 U/100 U
RNA Capping & Tailing	BE09019S/M	Poly(A) RNA Polymerase	100 U/500 U
RNA Analysis	BE05017S/M	RNase A	100 mg/1 g
	BE05018S/M	RNase T1	100 KU/5×100 KU
	BE05019S/M	Thermostable RNase T1	100 KU/5×100 KU
	BE05006S	RNase H	1000 U
	BE05011S	Thermostable RNase H	200 U



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